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# Thermodynamic calculations of the Au–Sc and Fe–Sc systems

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ABSTRACT

Thermodynamic optimization of the Au–Sc and Fe–Sc systems was carried out by means of the CALPHAD (CALculation of PHAse Diagram) method on the basis of the available experimental data in literature. Redlich–Kister polynomials were used to describe the excess Gibbs energy of solution phases, and all the compounds are treated as stoichiometric ones. The Au–Sc system was described thermodynamically for the first time, and the Fe–Sc system was re-optimized by considering the new experimental data about enthalpies of mixing of the liquid phase. A set of self-consistent parameters was obtained for each of these two binary systems, respectively.

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## 1. Introduction

Alloys with scandium addition have been considered to be a new generation of high-performance alloy structural materials for aerospace, ships and weapons [1], which is expected to have important advanced applications in many emerging industry fields. The addition also does improvements on many other kind of alloys for example to remove gases and other harmful impurities and reduce the occurrence of cracks during solidification of alloys. With the addition of a small amount of scandium, the tensile strength, elongation and hardness, as well as the corrosion and heat resistance of aluminum alloys can be significantly improved [2]. In addition, Fe–Sc alloys can be used as nanoglass material. Recently, Wang et al. [3] have discussed the plasticity of the scandium-based nanoglass. Moreover, Scandium addition into gold alloy can significantly strengthen alloy and raise the recrystallization temperature [4]. The phase equilibria and thermodynamic properties in the Au–Sc and Fe–Sc systems are of great importance for the designing of related new materials. However, thermodynamic description about the Au–Sc system is unavailable and the recently published thermodynamic description of the Fe–Sc system [5] cannot be reproduced right using the provided parameters.

The present work is devoted to performing thermodynamic modeling of the Au–Sc and Fe–Sc systems by considering the experimental data in literatures via CALPHAD approach [6].

## 2. Evaluation of the literature information

The available literature data on phase diagram and thermodynamic properties of the Au–Sc and Fe–Sc systems were critically reviewed in the present work and summarized in Tables 1 and 2.

### 2.1. The Au–Sc system

The experimental data on phase equilibrium of the Au–Sc system are limited. The Au–Sc phase diagram was originally reported by Palenzona et al. [7], in the compositional range from 10 to 80 at% Sc by means of differential thermal analysis (DTA) and X-ray diffraction (XRD). Rider et al. [8] investigated the solubility of Sc in (αAu) in the range from 0 to 10 at% Sc by XRD and reported that the largest solubility of Sc was up to 8.8 at%, and the eutectic temperature reached 1040 °C. Based on these data, Okamoto [9] reviewed the complete phase diagram. Generally, there are three solution phases (αAu, αSc and βSc phases), and six intermetallic compounds ($Au_4Sc$, $Au_3Sc$, $Au_2Sc$, AuSc, $AuSc_2$ and $Au_2Sc_7$) in the system as shown in Table 1.

Kleppa and Topor [10] and Fitzner et al. [11] investigated the enthalpies of mixing of the Au–Sc alloys at 1379 K and 1473 K by high-temperature reaction calorimetry (HTRC), respectively. Besides, Fitzner et al. [11] reported the standard enthalpy of formation of AuSc based on the observed heats of mixing and the average heat contents. Later, Wu et al. [12] calculated the formation enthalpies of disordered solid solutions and intermetallic compounds ($Au_4Sc$, $Au_3Sc$, $Au_2Sc$, AuSc and $AuSc_2$) at zero-temperature in the Au–Sc system with the modified analytical embedded atom method (EAM) [13,14] and the first-principles

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**Table 1**
Phases, structures and models of Au–Sc and Fe–Sc alloys.

| Phase | Structure type | Lattice constants (Å) | | | Ref. |
|---|---|---|---|---|---|
| | | *a* | *b* | *c* | |
| Au–Sc | | | | | |
| (αAu) | Cu | – | – | – | – |
| $Au_4Sc$ | $MoNi_4$ | 6.546 | – | 4.037 | [7] |
| | tI10-I4/m | 5.549 | – | 4.039 | [9,15] |
| $Au_3Sc$ | $TiCu_3$ | 5.800 | 4.823 | 5.003 | [7] |
| | oP8-Pmmn | | | | – |
| $Au_2Sc$ | $MoSi_2$ | 3.510 | – | 8.728 | [7] |
| | tI6-I4/mmm | 3.509 | – | 8.730 | [9,15] |
| AuSc | CsCl | 3.369 | – | – | [7] |
| | cP2-Pm3m | 3.370 | – | – | [9,15,16] |
| $AuSc_2$ | $PbCl_2$ | 6.470 | 4.667 | 8.358 | [7] |
| | oP12-Pnma | – | – | – | [17] |
| $Au_2Sc_7$ | $PbCl_2$ | 14.360 | 14.360 | 14.360 | [7] |
| | oI142-Immm | – | – | – | – |
| (αSc) | Mg | – | – | – | – |
| (βSc) | W | – | – | – | – |
| Fe–Sc | | | | | |
| (αFe) | W | – | – | – | – |
| (γFe) | Cu | – | – | – | – |
| (δFe) | W | – | – | – | – |
| $\beta Fe_2Sc$ | $MgNi_2$ | 4.974 | 4.974 | 16.29 | [32] |
| $\alpha Fe_2Sc$ | $MgZn_2$ | 4.948 | 4.948 | 8.077 | [33] |
| $\lambda Fe_2Sc$ | $Cu_2Mg$ | 7.039 | 7.039 | 7.039 | [32] |
| $Fe_6Sc_{29}$ | $Fe_6Sc_{29}$ | 14.361 | 14.361 | 14.361 | [32] |
| (αSc) | Mg | – | – | – | – |
| (βSc) | W | – | – | – | – |

calculations. All the available data for the Au–Sc system are summarized in Table 2.

### 2.2. The Fe–Sc system

The phase diagram was firstly experimentally investigated by Naumkin et al. [18]. Subsequently, Bodak et al. [19] reported the similar phase equilibria. Besides, Hellawell [20] measured the solubility of Sc in (αFe) and (δFe) in the range less than 1.5 at% Sc. The phase diagram of Fe–Sc system was assessed by Okamoto [21]. There are five solution phases (αFe, γFe, δFe, αSc and βSc phases), and four intermetallic compounds ($\alpha Fe_2Sc$, $\beta Fe_2Sc$, $\lambda Fe_2Sc$, and $Fe_6Sc_{29}$) in the system, as listed in Table 1.

It should be mentioned that Bodak et al. [19] reported the formula of $FeSc_7$ for the compound with the composition of $Fe_6Sc_{29}$. However, in most recently studies, the composition of $FeSc_7$ was redefined as $Fe_6Sc_{29}$ [22], and the related peritectoid reaction temperature was 865 °C. So this phase is modeling as $Fe_6Sc_{29}$ in the present work.

Partial and integral enthalpies of mixing of liquid Fe–Sc alloys were determined by Sudavtsova et al. [23] and Esin et al. [24] at 1870 K up to 11 at% Sc and 18 at% Sc, respectively. Most recently, the partial and integral mixing enthalpies of Fe–Sc melts at 1873 K over the entire composition range were determined by Sudavtsova et al. [25] by calorimetry. Enthalpies of formation for at the $Fe_2Sc$ and $Fe_6Sc_{29}$ phases at 1473 K have been experimentally determined using calorimetry by Selhaoui et al. [26] and Savchenkova et al. [27]. Besides, Burylev [28] and Boer et al. [29] calculated enthalpy formation of the $Fe_2Sc$ phase using the Miedema and Hildebrand-Mott models. All the experimental data for the Au–Sc system are summarized in Table 2.

The Fe–Sc system has been thermodynamically assessed by Liu et al. [5], as shown in Fig. 1. Obviously, the latest experimental data investigated by Sudavtsova et al. [25] was not considered in their work. There also exist some other calculation problems using the published parameters. One is that the calculated temperature for the eutectic reaction L↔(γFe)+$\beta Fe_2Sc$ was almost 30 °C lower, and the stable temperature range of $\lambda Fe_2Sc$ was much larger than reported ones in literature [18,19]. Another problem is that the calculated liquidus curve around 50 at% Sc was not fitted well and the enthalpies of mixing in their calculation were much lower comparing to the literature data [25]. All the problems mentioned before is due to one basic reason, the parameters of the liquid

**Table 2**
Summary of the literature data for the Au–Sc and Fe–Sc system.

| Type of data | Reference | Method | Quoted mode |
|---|---|---|---|
| Au–Sc | | | |
| Liquidus | | | |
| 10–80 at% Sc | [7] | DTA | ■ |
| Solubility | | | |
| Sc in (αAu) | [8] | XRD | ■ |
| Homogeneity of different compounds | | | |
| $Au_4Sc$ | [7] | XRD and MA | ■ |
| | [9,15] | XRD | □ |
| $Au_3Sc$ | [7] | XRD and MA | ■ |
| $Au_2Sc$ | [7] | XRD and MA | ■ |
| | [9,15] | XRD | □ |
| AuSc | [7] | XRD and MA | ■ |
| | [9,15] | XRD | □ |
| $AuSc_2$ | [7] | XRD and MA | ■ |
| $Au_2Sc_7$ | [7] | XRD and MA | ■ |
| Eutectic temperature | [7] | DTA | ■ |
| Peritectic temperature | [7] | DTA | ■ |
| Enthalpies of formation | [11] | High tem. calorimetry | ■ |
| | [12] | EAM | □ |
| | [12] | VASP | □ |
| | [11] | Miedema | + |
| Enthalpy of mixing of liquid | [10,11] | High tem. calorimetry | ■ |
| | [11] | Miedema | + |
| Fe–Sc | | | |
| Liquidus | | | |
| 0–100 at% Sc | [18] | DTA | ■ |
| 0–80 at% Sc | [19] | DTA | ■ |
| Solubility | | | |
| Sc in (αFe) and (δFe) | [20] | XRD, MA and DTA | ■ |
| Homogeneity of different compounds | | | |
| $\alpha Fe_2Sc$ | [33] | XRD | ■ |
| $\beta Fe_2Sc$ | [32] | XRD | ■ |
| $\lambda Fe_2Sc$ | [32] | XRD | ■ |
| $Fe_6Sc_{29}$ | [32] | XRD | ■ |
| Eutectic temperature | [18] | DTA | ■ |
| | [19] | DTA | ■ |
| Peritectic temperature | [18] | DTA | ■ |
| | [19] | DTA | ■ |
| Enthalpies of formation | [26] | High tem. calorimetry | ■ |
| | [27] | High tem. calorimetry | + |
| | [28] | Miedema | + |
| | [29] | Hildebrand-Mott models | + |
| Partial and integral enthalpy of mixing | [23] | High tem. calorimetry | ■ |
| | [24] | High tem. calorimetry | □ |
| | [25] | High tem. calorimetry | ■ |

DTA: differential thermal analysis; MA: metallographic analysis; XRD: X-ray diffraction; EAM: embedded atom method; VASP: Vienna Ab-initio simulation package.
Indicates whether the data are used or not used in the optimization procedure: ■, used; □, not used but considered as reliable data for checking the modeling; +, not used.

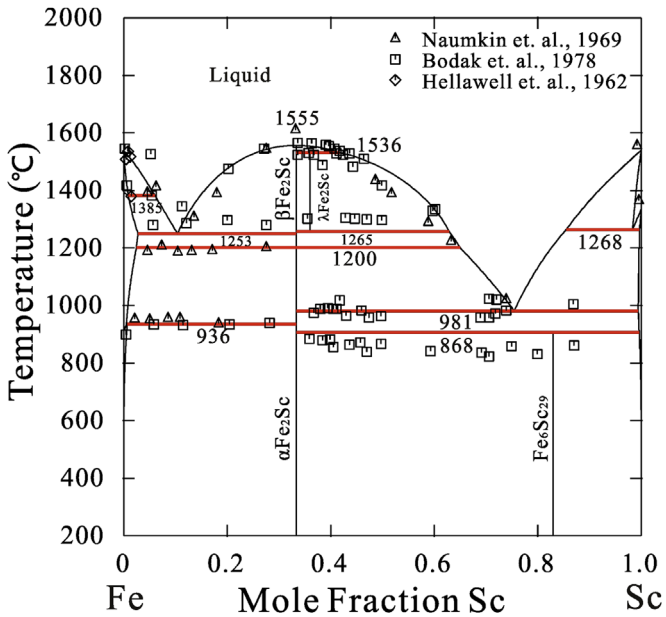


**Fig. 1.** The phase diagram of the Fe–Sc system calculated using the parameters of Liu et al. [5] comparing with the experimental results [18–20].

phase were inaccurate.

Recently, Sudavtsova et al. [25] plotted the liquidus curve for the Fe–Sc phase diagram according to the model of ideal associated solutions (IAS) [30,31] and their thermodynamic parameters. Based on the effect of the IAS model, not only the trend of the liquidus curve, but also their eutectic reaction points and temperatures differed from the calculation carried out by Liu et al. [5].

It is essential to re-optimize this system, since the accurate thermodynamic description is very necessary for the establishment of thermodynamic database and the development of advanced materials.

## 3. Thermodynamic modeling

### 3.1. Unary phases

The thermodynamic parameters of pure elements Au, Fe and Sc are taken from the SGTE-compilation by Dinsdale [34] described as follows:

$$G_i(T) - H_i^{SER} = A + B \times T + C \times T \times lnT + D \times T^2 + E \times T^{-1} + F \times T^3 + I \times T^7 + J \times T^{-9} \quad (1)$$

where $H_i^{SER}$ is the molar enthalpy of the pure element $i$ at 298.15 K and 1 bar in its standard element reference (SER) state, and T is the absolute temperature.

### 3.2. Solution phases

The liquid, (αAu), (αFe), (γFe), (δFe), (αSc) and (βSc) phases are modeled as completely disordered solution phases, and their molar Gibbs energies are described in the form of:

$$G^{\varphi} - H^{SER} = x_{Tm} \times {}^0G_{Tm}^{\varphi} + x_{Sc} \times {}^0G_{Sc}^{\varphi} + R \times T \times (x_{Tm} \times \ln x_{Tm} + x_{Sc} \times \ln x_{Sc}) + x_{Tm} \times x_{Sc} \times L_{Tm,Sc}^{\varphi} \quad (2)$$

where $x_{Tm}$ is the mole fraction of the element Tm (Tm=Au or Fe), and ${}^0G_{Tm}^{\varphi}$ is the Gibbs energy of element Tm under the state of φ. $L_{Tm,Sc}^{\varphi}$ stands for the interaction parameter in the form of the Redlich–Kister (R–K) polynomial [35]:

$$L_{Tm,Sc}^{\varphi} = \sum_{i=0}^{n} L_{Tm,Sc}^{i,\varphi} \times (x_{Tm} - x_{Sc})^i \quad (3)$$

where $L_{Tm,Sc}^{i,\varphi} = a_i + b_i T$ is the $i$th interaction parameter of the solution phase φ in the Au–Sc and Fe–Sc systems, and $a_i$ and $b_i$ are the parameters that should be optimized in the present work.

The equation about the Fe–Sc system in Eq. (4) is the magnetic contribution to the Gibbs energy. For the liquid phase, ${}^{mag}G^{\varphi}$ is equal to zero, and for other solution phases, the magnetic contribution from pure Fe to the Gibbs energy is given by the Hillert–Jarl–Inden model [36]:

$$G_m^{mag} = R \times T \times \ln(\beta^{\varphi}+1) \times g(\tau^{\varphi}) \quad (4)$$

where $\beta$ is the Bohr magnetic moment per mole atoms. As for the variable $\tau$, there exists one expression, $\tau = T/T_c^*$, $T_c^*$ is defined as the critical temperature for magnetic ordering (Curie or Neel temperature). The complete description of the function $g(\tau)$ is given in Ref. [36], as follows:

$$g(\tau^{\phi}) = 1 - \frac{\left[79\tau^{-1}/140p + 474/497\left(\frac{1}{p-1}\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right]}{D}, \text{ for } \tau \le 1 \quad (5)$$

$$g(\tau^{\phi}) = -\frac{\left(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500\right)}{D}, \text{ for } \tau > 1 \quad (6)$$

$$D = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right) \quad (7)$$

where the parameter $p$ depends on the structure, 0.40 for bcc structure and 0.28 for others.

### 3.3. Intermetallic phases

In these two systems, ten intermetallic compounds have been mentioned. $Au_4Sc$, $Au_3Sc$, $Au_2Sc$, AuSc, $AuSc_2$ and $Au_2Sc_7$ phases in the Au–Sc system, and $\alpha Fe_2Sc$, $\beta Fe_2Sc$, $\lambda Fe_2Sc$, and $Fe_6Sc_{29}$ phases in the Fe–Sc system.

The Gibbs energy per mole atoms of $Tm_xSc_{(1-x)}$ compound is described as follow:

$${}^0G_m^{Tm_xSc_{1-x}} - x \times H_{Tm}^{SER} - (1-x) \times H_{Sc}^{SER} = a + b \times T + x \times {}^0G_{TM}^{SER} + (1-x) \times {}^0G_{Sc}^{Hcp} \quad (8)$$

In which, a and b are the parameters that needed to be optimized in this work.

## 4. Optimization procedure and results

The optimization of the thermodynamic parameters for these two systems was carried out by using the PARROT module of the Thermo-Calc software [37]. It was started with the liquid phase by using the experimental data of partial and integral enthalpies of mixing. After that, the intermetallic compounds and other solution phases were taken into consideration one by one. All the intermetallic phases were treated as stoichiometric ones, and the experimentally measured enthalpies of formation for the intermetallic phases were used as good starting values for the

**Table 3**
Thermodynamic parameters in the Au–Sc and Fe–Sc system.

| Phase | Parameters |
|---|---|
| Au–Sc | |
| Liquid | ${}^{0}L^{Liquid}_{Au,Sc} = -229326.82 + 8.12 \times T$ |
| | ${}^{1}L^{Liquid}_{Au,Sc} = -37303.77 - 14.29 \times T$ |
| | ${}^{2}L^{Liquid}_{Au,Sc} = 3441.85 + 2.39 \times T$ |
| Fcc | ${}^{0}L^{Fcc}_{Au,Sc} = -175520.06 - 22.02 \times T$ |
| | ${}^{1}L^{Fcc}_{Au,Sc} = -65519.78$ |
| $Au_4Sc$ | ${}^{0}G^{Au_4Sc}_{Au,Sc} = 0.80 \times {}^{0}G^{Bcc}_{Au} + 0.20 \times {}^{0}G^{Hcp}_{Sc} - 54000 + 4.65 \times T$ |
| $Au_3Sc$ | ${}^{0}G^{Au_3Sc}_{Au,Sc} = 0.75 \times {}^{0}G^{Bcc}_{Au} + 0.25 \times {}^{0}G^{Hcp}_{Sc} - 60000 + 3.25 \times T$ |
| $Au_2Sc$ | ${}^{0}G^{Au_2Sc}_{Au,Sc} = 0.67 \times {}^{0}G^{Bcc}_{Au} + 0.33 \times {}^{0}G^{Hcp}_{Sc} - 66986.43 + 0.54 \times T$ |
| AuSc | ${}^{0}G^{AuSc}_{Au,Sc} = 0.50 \times {}^{0}G^{Bcc}_{Au} + 0.50 \times {}^{0}G^{Hcp}_{Sc} - 76100 + 3.70 \times T$ |
| $AuSc_2$ | ${}^{0}G^{AuSc2}_{Au,Sc} = 0.33 \times {}^{0}G^{Bcc}_{Au} + 0.67 \times {}^{0}G^{Hcp}_{Sc} - 63000 + 7.72 \times T$ |
| $Au_2Sc_7$ | ${}^{0}G^{Au_2Sc_7}_{Au,Sc} = 0.22 \times {}^{0}G^{Bcc}_{Au} + 0.78 \times {}^{0}G^{Hcp}_{Sc} - 53000 + 12.50 \times T$ |
| Fe–Sc | |
| Liquid | ${}^{0}L^{Liquid}_{Fe,Sc} = -49300 - 14.2 \times T$ |
| | ${}^{1}L^{Liquid}_{Fe,Sc} = -7000 - 5.3 \times T$ |
| | ${}^{2}L^{Liquid}_{Fe,Sc} = 8000$ |
| Fcc | ${}^{0}L^{Fcc}_{Fe,Sc} = -57180 + 4.7 \times T$ |
| Bcc | ${}^{0}L^{Bcc}_{Fe,Sc} = -52280 + 13.8 \times T$ |
| | ${}^{1}L^{Bcc}_{Fe,Sc} = -16220$ |
| Hcp | ${}^{0}L^{Hcp}_{Fe,Sc} = 3700$ |
| $\alpha Fe_2Sc$ | ${}^{0}G^{\alpha Fe_2Sc}_{Fe,Sc} = 0.67 \times {}^{0}G^{Bcc}_{Fe} + 0.33 \times {}^{0}G^{Hcp}_{Sc} - 41266 + 7.3 \times T$ |
| | $T_C^{\alpha Fe_2Sc} = 542$ |
| | $\beta^{\alpha Fe_2Sc} = 1.37$ |
| $\beta Fe_2Sc$ | ${}^{0}G^{\beta Fe_2Sc}_{Fe,Sc} = 0.67 \times {}^{0}G^{Bcc}_{Fe} + 0.33 \times {}^{0}G^{Hcp}_{Sc} - 41070 + 7.2 \times T$ |
| $\lambda Fe_2Sc$ | ${}^{0}G^{\lambda Fe_2Sc}_{Fe,Sc} = 0.64 \times {}^{0}G^{Bcc}_{Fe} + 0.36 \times {}^{0}G^{Hcp}_{Sc} - 33195 + 2.6 \times T$ |
| $Fe6Sc_{29}$ | ${}^{0}G^{Fe_6Sc_{29}}_{Fe,Sc} = 0.17 \times {}^{0}G^{Bcc}_{Fe} + 0.83 \times {}^{0}G^{Hcp}_{Sc} - 12410 + 3.5 \times T$ |

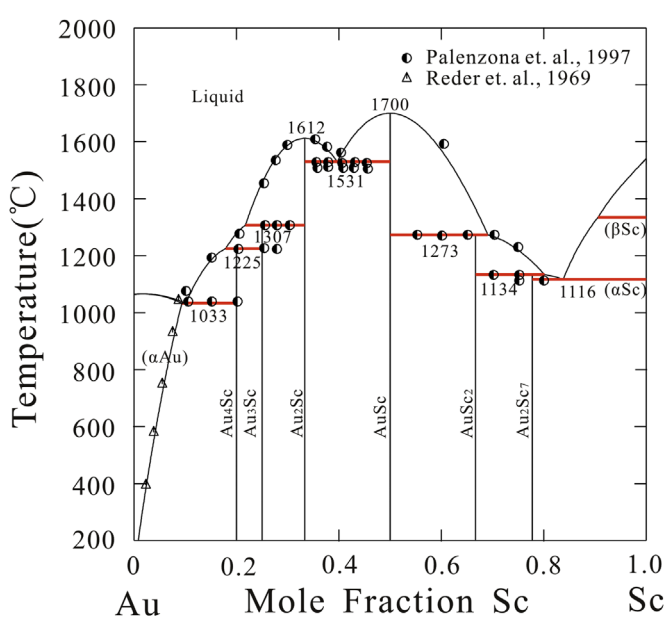


**Fig. 2.** The calculated phase diagram of the Au–Sc system compared with the experimental data [7,8].

**Table 4**
Summary of the invariant equilibria in the Au–Sc and Fe–Sc system.

| Reaction | T (K) | | Type | Phase | Composition, at% Sc | |
|---|---|---|---|---|---|---|
| | Exp. [7,21] | Calc. | | | Exp. [7,8,21] | Calc. |
| Au–Sc | | | | | | |
| $L \leftrightarrow Au_4Sc + (\alpha Au)$ | 1313 | 1306 | Eutectoid | L | 8.8∼14 | 10.1 |
| | | | | $Au_4Sc$ | 20 | 20 |
| | | | | (αAu) | 0∼8.8 | 9.4 |
| $L + Au_3Sc \leftrightarrow Au_4Sc$ | 1498 | 1498 | Peritectic | L | 17.2 | 17.9 |
| | | | | $Au_3Sc$ | 25 | 25 |
| | | | | $Au_4Sc$ | 20 | 20 |
| $L + Au_2Sc \leftrightarrow Au_3Sc$ | 1578 | 1580 | Peritectic | L | 22.8 | 21.8 |
| | | | | $Au_2Sc$ | 33.3 | 33.3 |
| | | | | $Au_3Sc$ | 25 | 25 |
| $L \leftrightarrow Au_2Sc$ | 1888 | 1885 | Congruent | L | 33.3 | 33.3 |
| | | | | $Au_2Sc$ | 33.3 | 33.3 |
| $L \leftrightarrow AuSc + Au_2Sc$ | 1798 | 1804 | Eutectoid | L | 39.5 | 39.7 |
| | | | | AuSc | 50 | 50 |
| | | | | $Au_2Sc$ | 33.3 | 33.3 |
| $L \leftrightarrow AuSc$ | 1973 | 1973 | Congruent | L | 50 | 50 |
| | | | | AuSc | 50 | 50 |
| $L + AuSc \leftrightarrow AuSc_2$ | 1548 | 1546 | Peritectic | L | 70 | 69.1 |
| | | | | AuSc | 50 | 50 |
| | | | | $AuSc_2$ | 66.7 | 66.7 |
| $L + AuSc_2 \leftrightarrow Au_2Sc_7$ | 1408 | 1407 | Peritectic | L | 81.2 | 80.1 |
| | | | | $AuSc_2$ | 66.7 | 66.7 |
| | | | | $Au_2Sc_7$ | 77.8 | 77.8 |
| $L \leftrightarrow Au_2Sc_7 + (\alpha Sc)$ | 1388 | 1389 | Eutectoid | L | 82.2 | 84.5 |
| | | | | $Au_2Sc_7$ | 77.8 | 77.8 |
| | | | | (αSc) | 98∼100 | 100 |
| Fe–Sc | | | | | | |
| $L \leftrightarrow (\delta Fe)$ | 1811 | 1811 | Melting | L | 100 | 100 |
| | | | | (δFe) | 100 | 100 |
| $(\delta Fe) \leftrightarrow (\gamma Fe)$ | 1668 | 1668 | Allotropic | (δFe) | 100 | 100 |
| | | | | (γFe) | 100 | 100 |
| $(\delta Fe) \leftrightarrow (\gamma Fe) + L$ | 1653 | 1653 | Peritectic | (δFe) | – | – |
| | | | | (γFe) | – | 1.07 |
| | | | | L | – | 6.73 |
| $(\gamma Fe) \leftrightarrow (\alpha Fe)$ | 1185 | 1185 | Allotropic | (γFe) | 100 | 100 |
| | | | | (αFe) | 100 | 100 |
| $(\gamma Fe) + \alpha Fe_2Sc \leftrightarrow (\alpha Fe)$ | 1208 | 1208 | Peritectoid | (γFe) | – | 0.59 |
| | | | | $\alpha Fe_2Sc$ | 33.3 | 33.3 |
| | | | | (αFe) | – | 0.75 |
| $L \leftrightarrow (\gamma Fe) + \beta Fe_2Sc$ | 1553 | 1553 | Eutectic | L | 11.5 | 10.4 |
| | | | | (γFe) | – | 2.58 |
| | | | | $\beta Fe_2Sc$ | 33.3 | 33.3 |
| $L \leftrightarrow \beta Fe_2Sc$ | 1838 | 1838 | Congruent | L | 33.3 | 33.3 |
| | | | | $\beta Fe_2Sc$ | 33.3 | 33.3 |
| $\beta Fe_2Sc \leftrightarrow \alpha Fe_2Sc$ | 1473 | 1473 | Poly-morphic | $\beta Fe_2Sc$ | 33.3 | 33.3 |
| | | | | $\alpha Fe_2Sc$ | 33.3 | 33.3 |
| $L + \beta Fe_2Sc \leftrightarrow \lambda Fe_2Sc$ | 1798 | 1798 | Peritectic | L | – | 42.0 |
| | | | | $\beta Fe_2Sc$ | 33.3 | 33.3 |
| | | | | $\lambda Fe_2Sc$ | 35.7 | 36.0 |
| $\lambda Fe_2Sc \leftrightarrow \beta Fe_2Sc + L$ | 1568 | 1568 | Peritectic | $\lambda Fe_2Sc$ | 35.7 | 36.0 |
| | | | | $\beta Fe_2Sc$ | 33.3 | 33.3 |
| | | | | L | 39.0 | 57.9 |
| $L \leftrightarrow \alpha Fe_2Sc + (\alpha Sc)$ | 1248 | 1248 | Eutectic | L | 74.0 | 73.6 |
| | | | | $\alpha Fe_2Sc$ | 33.3 | 33.3 |
| | | | | (αSc) | – | 99.5 |
| $\alpha Fe_2Sc + (\alpha Sc) \leftrightarrow Fe_6Sc_{29}$ | 1138 | 1139 | Peritectoid | $\alpha Fe_2Sc$ | 33.3 | 33.3 |
| | | | | (αSc) | – | 99.7 |
| | | | | $Fe_6Sc_{29}$ | 87.5 | 83.0 |
| $L \leftrightarrow (\beta Sc)$ | 1814 | 1814 | Melting | L | 0 | 0 |
| | | | | (βSc) | 0 | 0 |
| $(\beta Sc) \leftrightarrow (\alpha Sc)$ | 1610 | 1610 | Poly-morphic | (βSc) | 0 | 0 |
| | | | | (αSc) | 0 | 0 |
| $(\beta Sc) \leftrightarrow L + (\alpha Sc)$ | 1544 | 1544 | Peritectic | (βSc) | – | 98.6 |
| | | | | L | – | 85.3 |
| | | | | (αSc) | – | 99.7 |

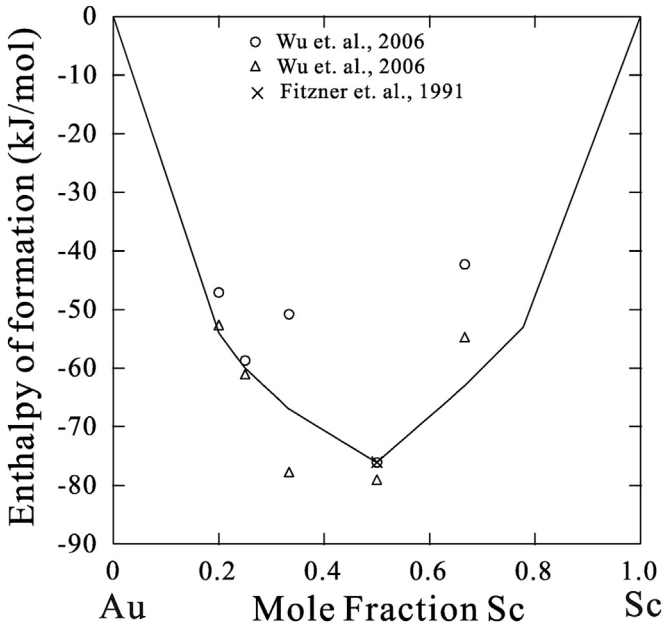


**Fig. 3.** Calculated Enthalpy of formation in the Au–Sc system with experiment data from Fitzner et al. [11] and Calculation from Wu et al. [12] by using EAM and the first-principles calculations.

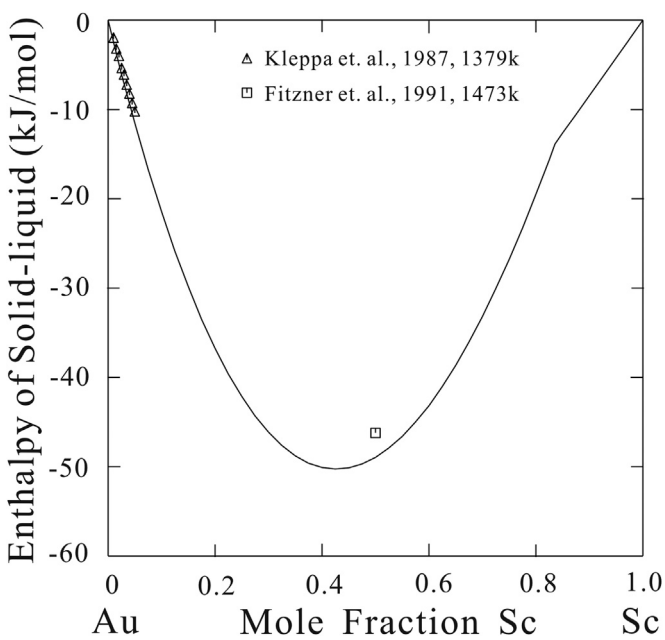


**Fig. 4.** Calculated solid-liquid enthalpies of mixing of Au(l)+Sc(s) at 1379 K and 1473 K with experiment data [10,11].

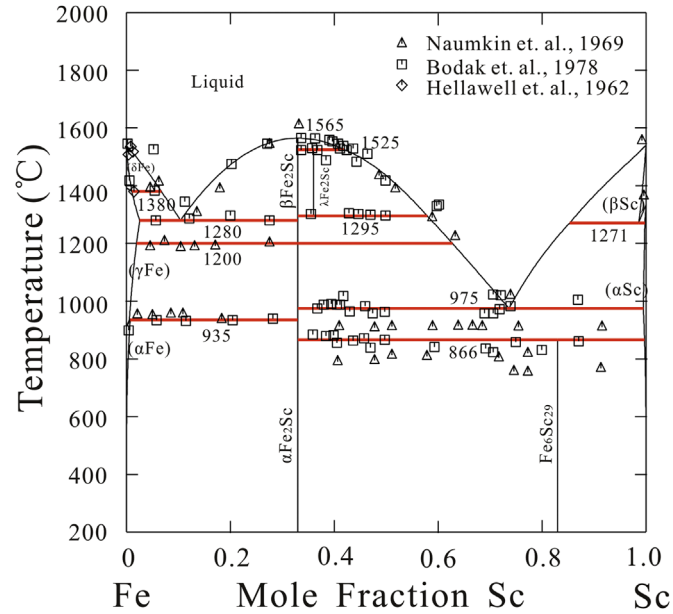


**Fig. 5.** The calculated phase diagram of the Fe–Sc system compared with the experimental data [18–20].

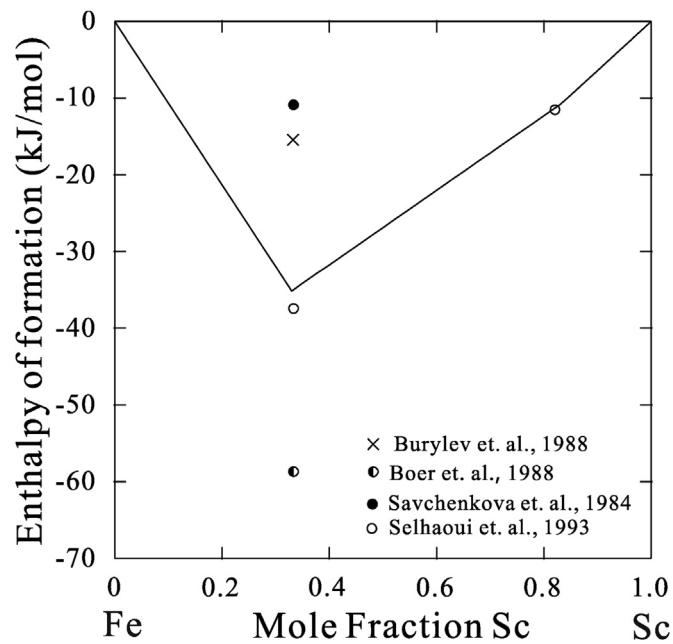


**Fig. 6.** Calculated enthalpies of formation in the Fe–Sc system with experiment data evaluated in Selhaoui et al. [26], Savchenkova et al. [27] by calorimetry and Calculated in Burylev et al. [28], Boer et al. [29] using the Miedema and Hildebrand-Mott models.

parameters a in Eq. (8). Parameters $a_i$ and $b_i$ for the solution phases were introduced to describe the solubility ranges. At last all parameters for different phases were optimized simultaneously to achieve a globally self-consistent thermodynamic description. The finally obtained thermodynamic parameters are listed in Table 3.

### 4.1. The Au–Sc system

The calculated Au–Sc phase diagram using the present parameters is shown in Fig. 2, and compared with the experimental data taken from Refs. [7,8]. Table 4 lists the invariant equilibria in the Au–Sc system. The experimental data of Sc-rich region is limited. Thus the solubility boundary of Au in solution phase (αSc) and (βSc) is taken to be zero in the optimization. In the range from 50 to 65 at% Au, Palenzona et al. [7] observed a thermal effect at 1510 °C. Since this kind of behavior was not able to explain clearly, it is not taken into consideration in the present work.

Fig. 3 shows the calculated enthalpy of formation in the Au–Sc system at 298 K compared with the literature data [11,12]. From the figure, we can see that the enthalpy of formation of the AuSc phase fits well to the experimental data, and the trends of other intermetallic phases are also reasonable to some extent.

Fig. 4 shows the calculated solid-liquid enthalpies of mixing of Au (l)+Sc (s) at 1379 K and 1473 K compared with the experimental data [10,11], and the calculated data reasonable. As a result of dropping solid scandium into liquid gold in their experiments,

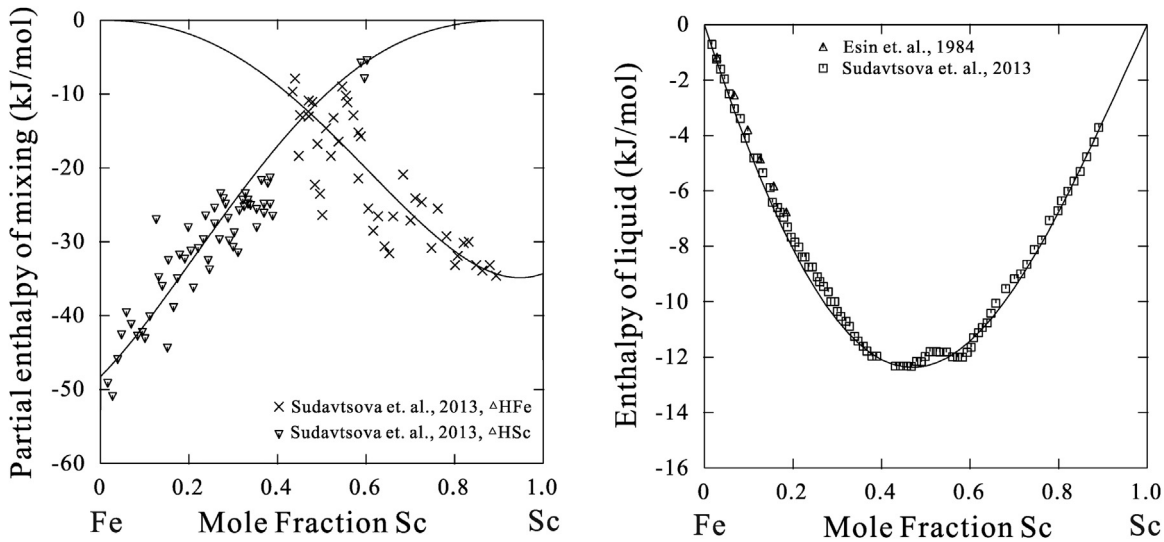


**Fig. 7.** Calculated mixing integral and partial enthalpies of liquid phase at 1873 K, (a) with experiment data from Esin et al. [24] and Sudavtsova et al. [25], (b) with experiment data determined by Sudavtsova et al. [23,25] at 1870 K up to 11 at% Sc and by Esin et al. [24] up to 18 at% Sc.

the reference state of scandium in this work is taken as solid (αSc). In the present work, temperature dependence of enthalpies is not taken into consideration due to the limited and dispersed experimental data below 1473 K.

### 4.2. The Fe–Sc system

The calculated Fe–Sc phase diagram based on the presently obtained thermodynamic parameters is shown in Fig. 5, and compared to the experimental data [18–20]. Table 4 shows the invariant equilibria in the Fe–Sc system.

In this work, the stable temperature range of $\lambda Fe_2Sc$ locates between 1295 °C to 1525 °C, which fits well with the literature information [21]. In addition, the latest experimental data of enthalpy of mixing investigated by Sudavtsova et al. [25] are considered in the optimization of the liquid phase, which leads to the much better curvature of the liquidus, in comparison with the previous assessment [5]. According to the present calculations, the calculated temperatures for the eutectic reactions $L \leftrightarrow (\gamma Fe) + \beta Fe_2Sc$ and $L \leftrightarrow \alpha Fe_2Sc + (\alpha Sc)$ are 1280 °C and 975 °C, which are different from the Fe–Sc phase diagram plotted according to the IAS model [25].

Fig. 6 presents the calculated enthalpies of formation at 298 K in the Fe–Sc system compared with the literature data [26–29]. Several different literature data about the enthalpy of formation of $Fe_2Sc$ located in a really large energy region. The data from the work of Selhaoui et al. [26] and Boer et al. [29] were more exothermic than those reported in others [27,28]. The present calculation can reproduce the data from Selhaoui et al. [26] reasonably.

Fig. 7 shows the calculated integral and partial molar enthalpies of mixing for liquid at 1873 K compared with the experimental data [23–25]. The experimental data reported by Sudavtsova et al. [25] for partial enthalpies of mixing distributes pretty discrete, while the calculated result passes through the central region of these dots, and goes pretty well.

## 5. Summary

All the literature phase diagram data and thermodynamic data of the Au–Sc and Fe–Sc binary systems on the literature were critically evaluated in this work. A set of thermodynamic parameters for each of the Au–Sc and Fe–Sc systems was obtained based on the CALPHAD method, respectively. The comparison between calculations and experiments shows that the calculations can reproduce the experimental data reasonably.

## Acknowledgements

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2016.07.002.

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